



Universitas
Esa Unggul

Smart, Creative and Entrepreneurial

PTI 412-Internet Of Things

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SESI 4

Konsep Dasar Service Discovery



Service Discovery Protocol for the IoT Ecosystem

*CSI 421-Internet Of Things
Universitas Esa Unggul*

- *Service Discovery Protocol for the IoT Ecosystem*
- **Semua Bahan mengacu kepada buku : The Internet of Things: Enabling Technologies, Platforms, and Use Cases [Pethuru Raj, Anupama C. Raman]**

Dasar Service Discovery

Dasar Service Discovery

Service Discovery adalah proses yang memungkinkan perangkat dalam jaringan untuk secara otomatis menemukan dan berinteraksi dengan layanan atau sumber daya. Ini penting untuk memfasilitasi komunikasi dan interaksi antar perangkat, terutama di lingkungan yang dinamis seperti IoT.



Peran dalam IoT

Dalam konteks Internet of Things (IoT), Service Discovery berperan krusial dengan memungkinkan perangkat untuk terhubung satu sama lain tanpa konfigurasi yang kompleks. Ini mendukung plug and play, mengurangi kebutuhan untuk pengaturan manual, dan meningkatkan interoperabilitas antar berbagai perangkat dan platform.

Mekanisme Discovery

Mekanisme Discovery mencakup berbagai teknik dan protokol untuk mendeteksi layanan dalam jaringan. Ini bisa bersifat aktif, di mana perangkat secara proaktif mencari layanan, atau pasif, di mana perangkat mendengarkan pengumuman dari layanan yang tersedia. Contoh teknik ini termasuk menggunakan multicast untuk mempercepat proses deteksi dan mengurangi latensi.

Protokol dan Kriteria

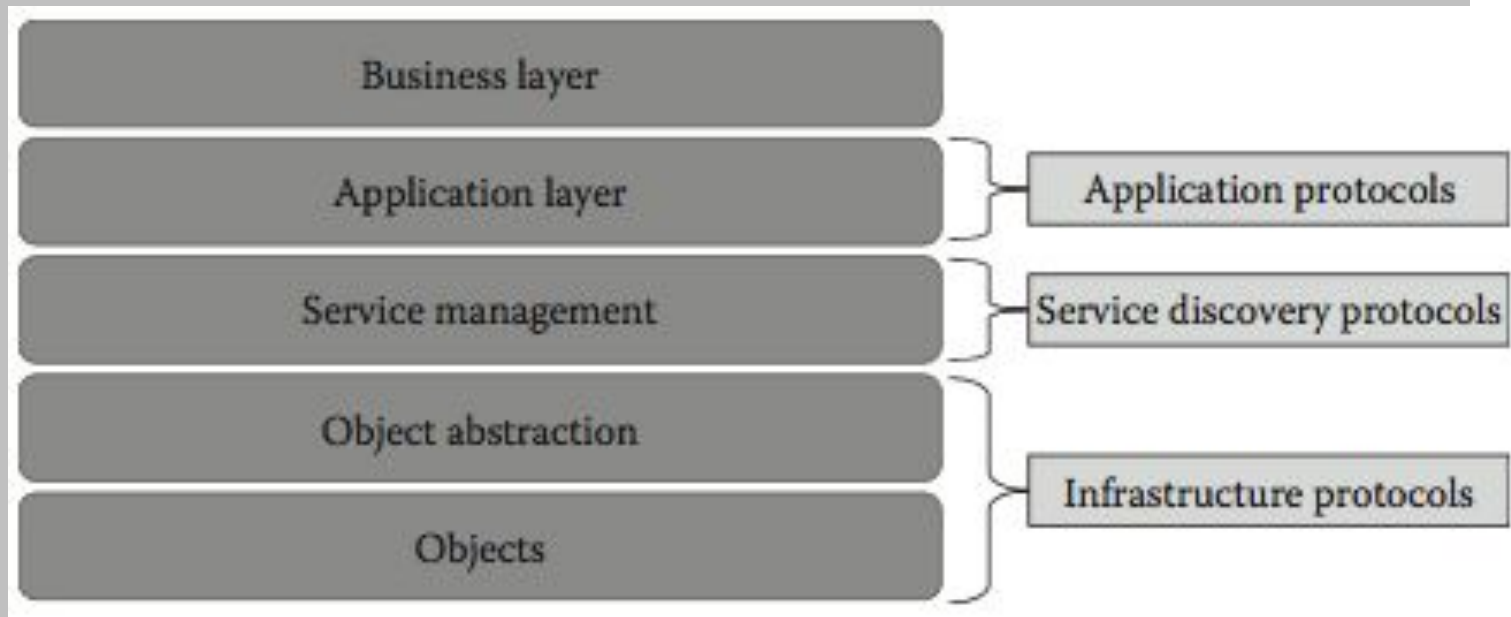
Protokol Populer IoT

Protokol populer dalam IoT termasuk

- *mDNS (Multicast Domain Name System) untuk jaringan lokal,*
- *SSDP (Simple Service Discovery Protocol) untuk menemukan perangkat secara otomatis di UPnP (Universal Plug and Play),*
 - *dan CoAP (Constrained Application Protocol) untuk perangkat yang terbatas sumber daya.*

Protokol-protokol ini dirancang untuk mendukung berbagai kebutuhan komunikasi, seperti penemuan perangkat dalam jaringan pintar dan komunikasi antar sensor.

Protocol architecture of IoT

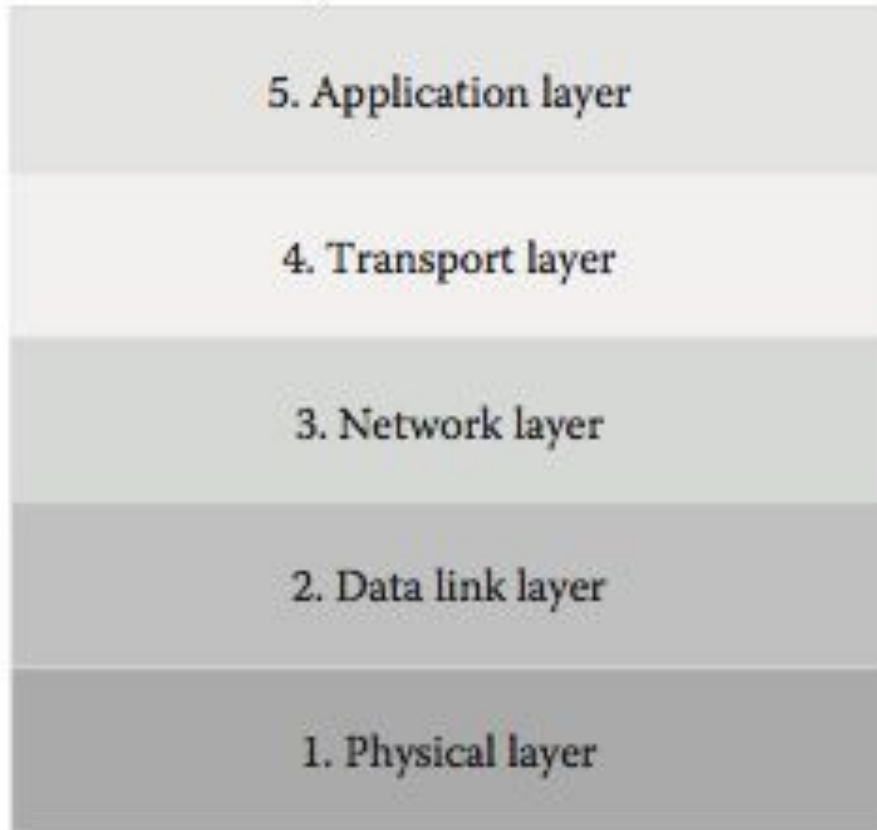


Evaluasi Protokol Discovery

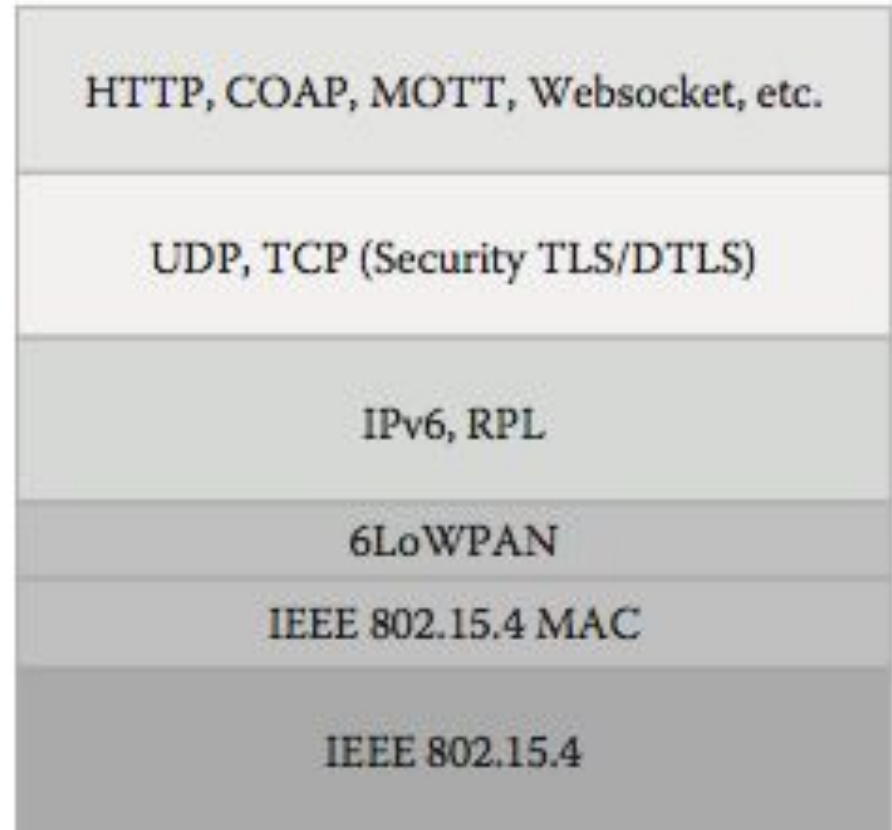
Kriteria evaluasi untuk protokol Discovery mencakup skalabilitas, konsumsi daya, dan kompatibilitas antar perangkat. Pertimbangan keamanan dan privasi juga sangat penting, terutama dalam konteks IoT, di mana data sensitif dapat dipertaruhkan. Latensi penemuan dan topologi jaringan juga perlu diperhitungkan.

Protocol stack of 6LoWPAN

Simplified OSI model



6LoWPAN stack



Arah Perkembangan Service Discovery

Tantangan utama dalam Service Discovery termasuk heterogenitas perangkat dan protokol yang berbeda, masalah konektivitas, serta isu keamanan dan autentikasi layanan. Interoperabilitas antar vendor juga menjadi kendala, karena berbagai produsen mungkin menerapkan solusi yang berbeda untuk penemuan layanan.

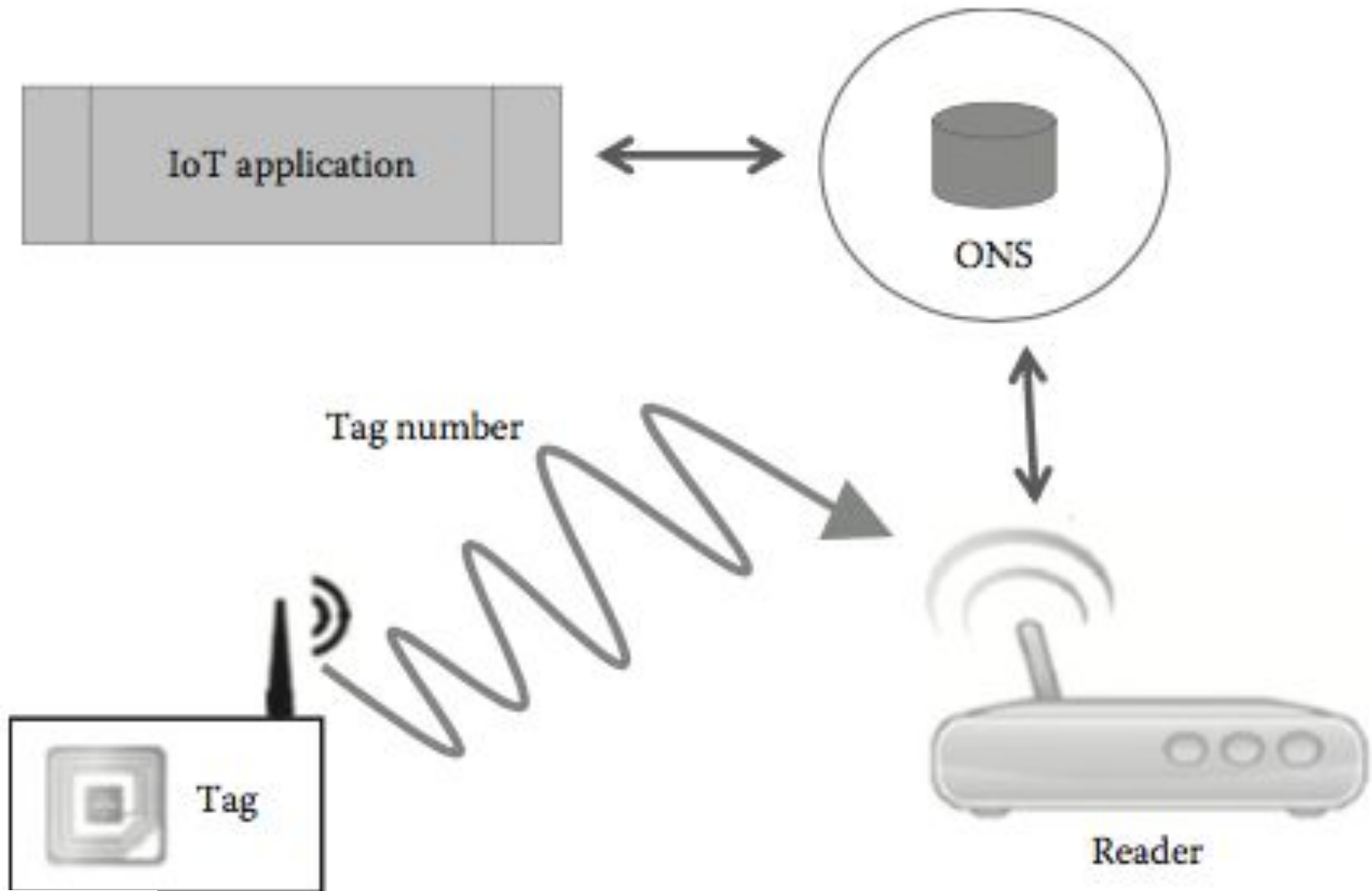
Studi Kasus Implementasi

Smart Home (HomeKit, Google Home)

Smart City (Sensor discovery)

Industrial IoT (gateway discovery)

Components of RFID system.



Video of RFID system.



Kesimpulan

Service Discovery sangat penting dalam mengoptimalkan komunikasi antar perangkat IoT. Dengan memahami mekanisme, protokol, dan tantangan terkait, kita dapat merancang solusi yang lebih efektif dan efisien untuk kebutuhan jaringan masa depan.

Terima Kasih

INFRASTRUCTURE PROTOCOLS

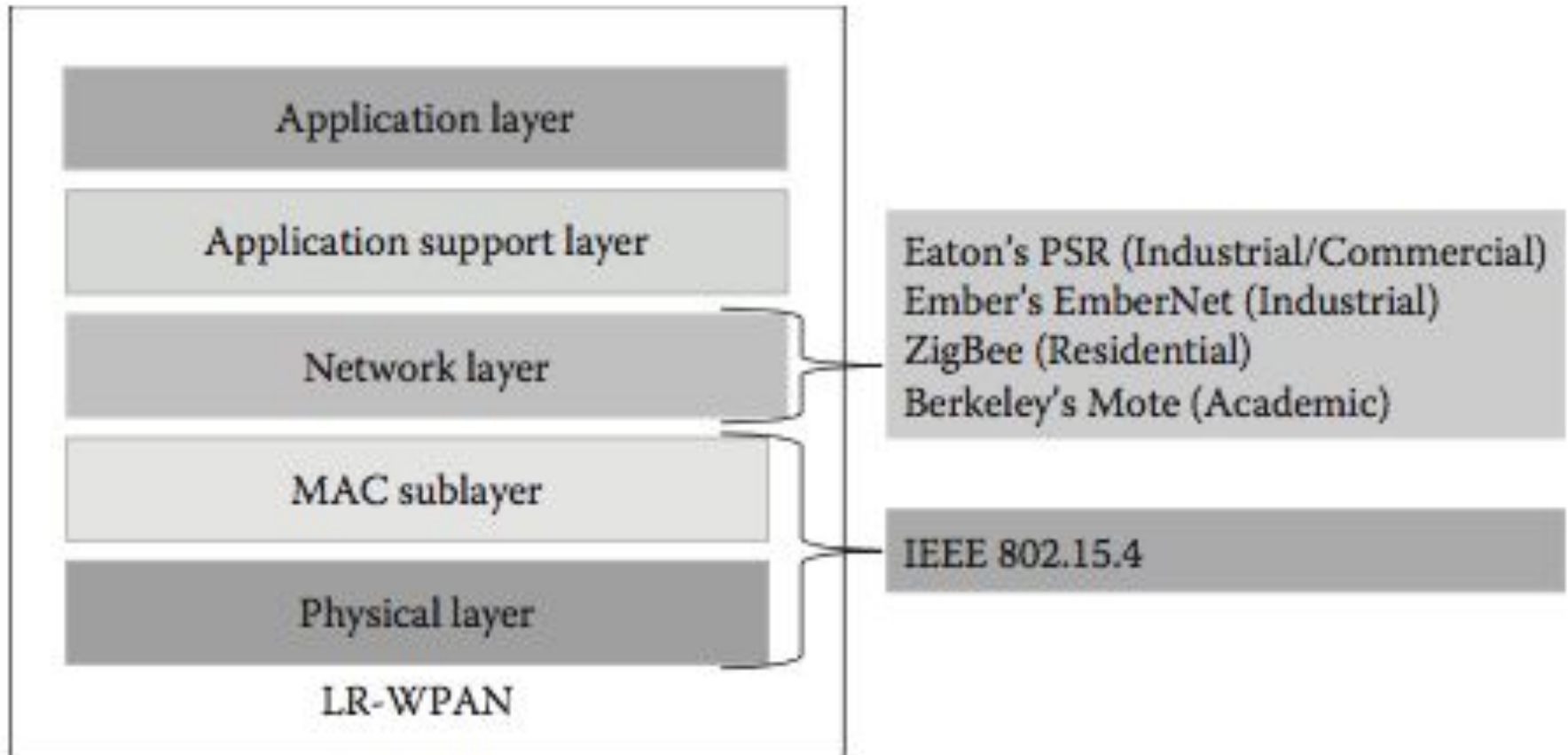
Categorization of IoT protocols

Application protocols		DDS	CoAP	AMQP	MQTT	MQTT-SN	XMPP	HTTP REST
Service discovery		mDNS				DNS-SD		
Infrastructure protocols	Routing protocol	RPL						
	Network layer	6LoWPAN				IPv4/IPv6		
	Link layer	IEEE 802.15.4						
	Physical/ device layer	LTE-A	EPCglobal	IEEE 802.15.4		Z-Wave		

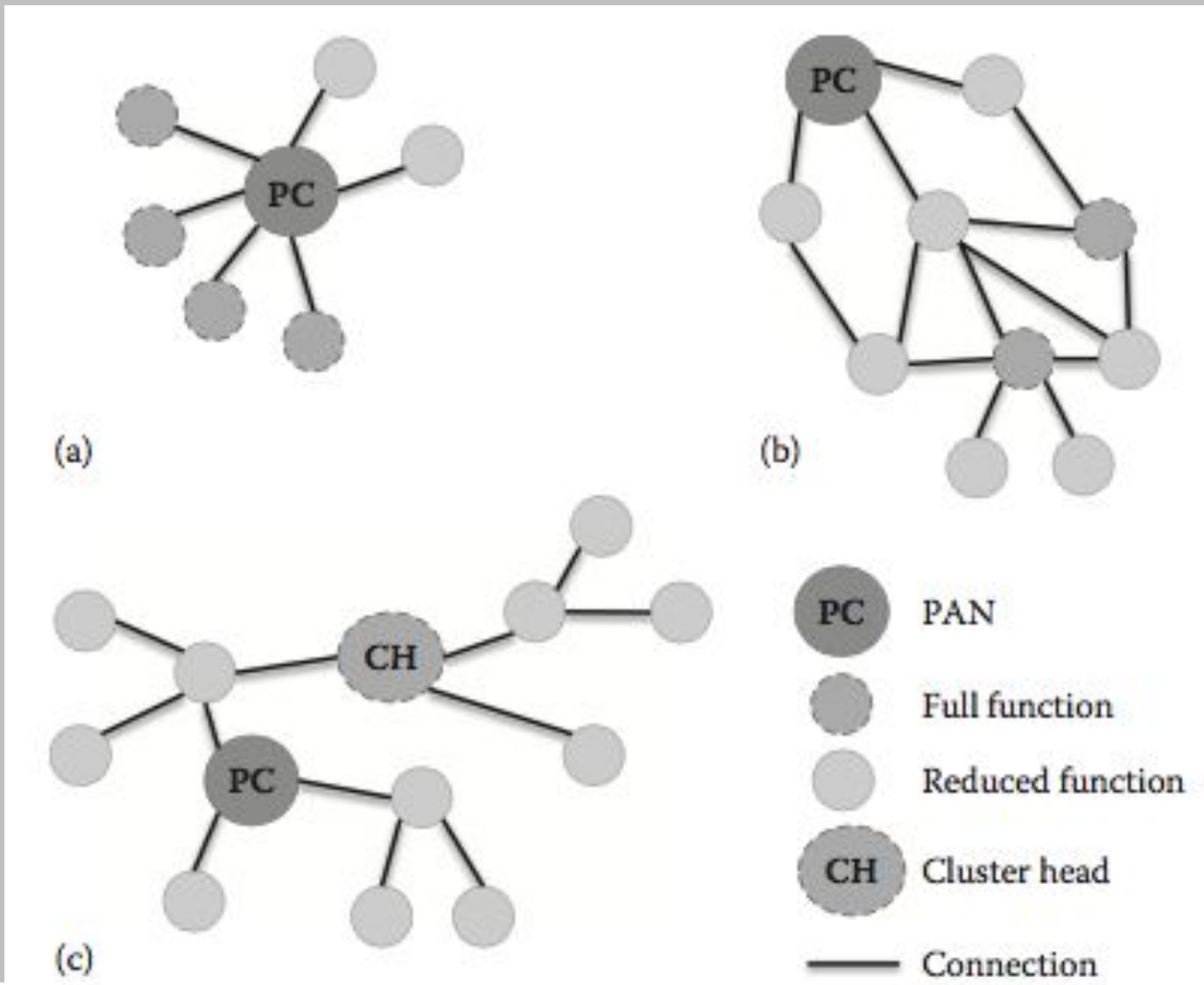
DODAG control messages

<i>Serial Number</i>	<i>Name of the Message</i>	<i>Description</i>
1	DODAG information object (DIO)	This message is used to keep the current rank (level) of the node, determine the distance of each node to the root based on some specific metrics, and choose the preferred parent path.
2	Destination advertisement object (DAO)	This message is used to unicast destination information toward selected parents of a node. This control message helps RPL to maintain upward and downward traffic.
3	DODAG information solicitation (DIS)	This message is used by a specific node in order to acquire DIO messages from another reachable adjacent node.
4	DAO acknowledgment (DAO-ACK)	This message is used as a response to a DAO message and is sent by a DAO recipient node like a DAO parent or DODAG root.

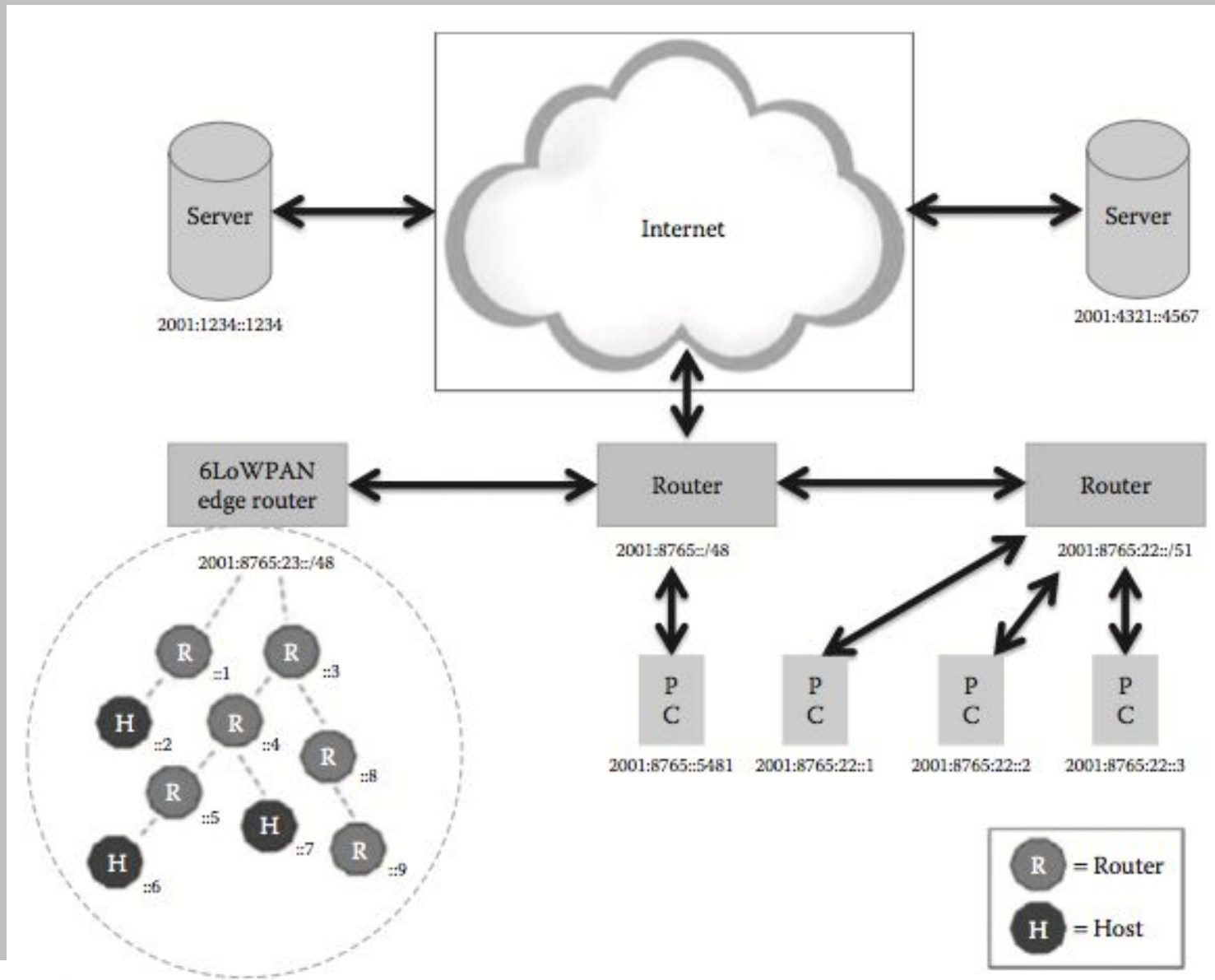
Architecture of IEEE 802.15.4.



(a–c) Different types of star topologies



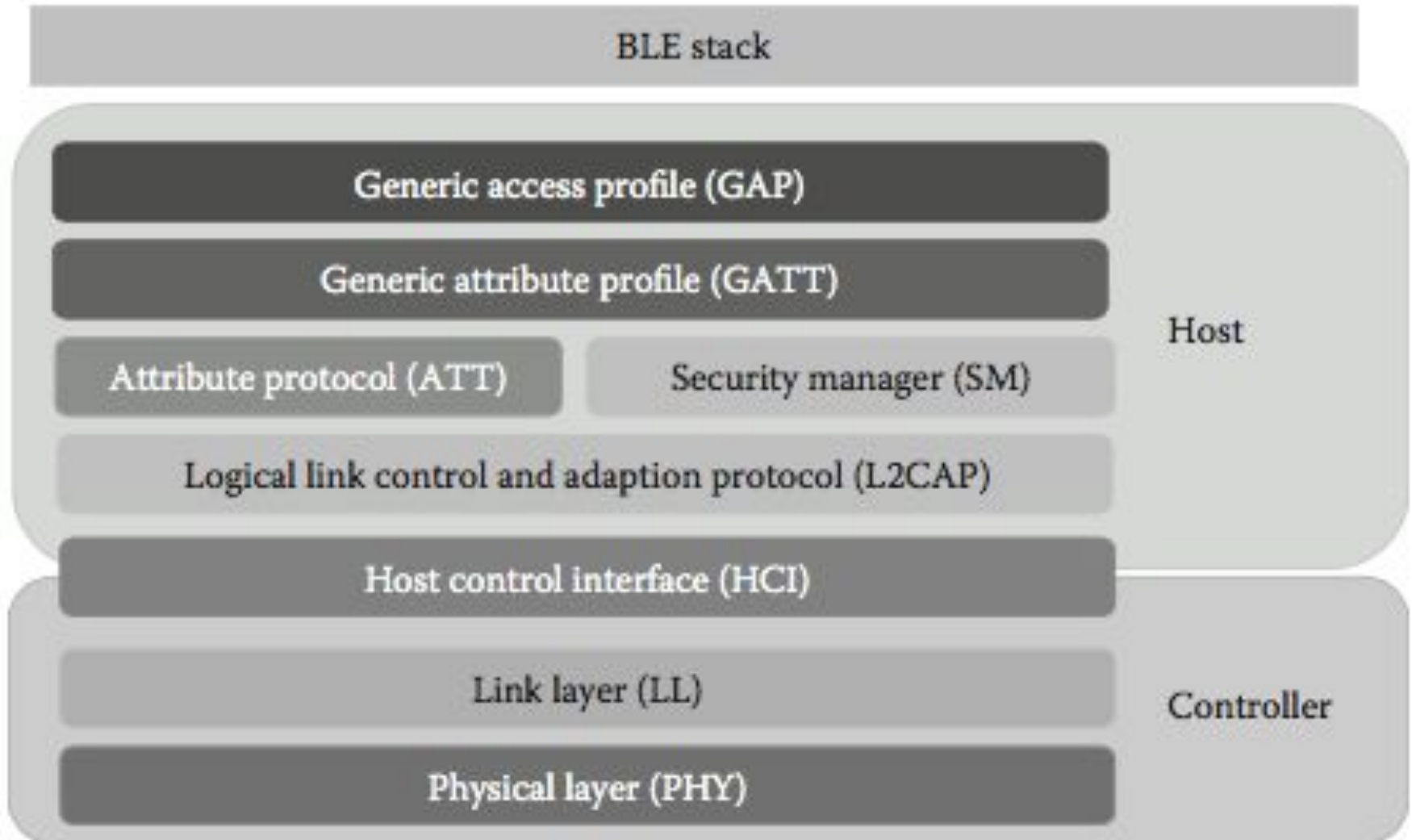
Network architecture of 6LoWPAN



Bluetooth Low Energy

- Bluetooth low energy (BLE) was started as part of the Bluetooth 4.0 core specification.
- BLE uses short-range radio with minimum power and operates for a long time.
- *Physical layer*: is layer receives and transmits data bits.
Link layer: Following are the functions performed by the link layer:
 - – Media access control
 - Error control
 - Connection establishment – Flow control

Protocol stack of BLE



Long Term Evolution-Advanced

- Long term evolution-advanced (LTE) also referred to as 4G LTE is a standard for wireless mobile network, and it provides high speed data transfer rates for wireless networks.
- It will provide 50 times performance improvement for existing wireless networks.
- LTE broadcast is a single- frequency network (SFN) that operates in a broadcast mode.

Different Types of EPC Tags

<i>EPC</i>	<i>Description</i>	<i>Tag Type</i>	<i>Functionality</i>
0	Read only	Passive	Write once and read many times
1	Write once and read only	Passive	Write once and read many times
2	Read or write	Passive	Read or write many times
3	Read or write	Semipassive	Attached within sensor
4	Read or write	Active	Attached within sensor While providing a radio wave field to communicate with the reader

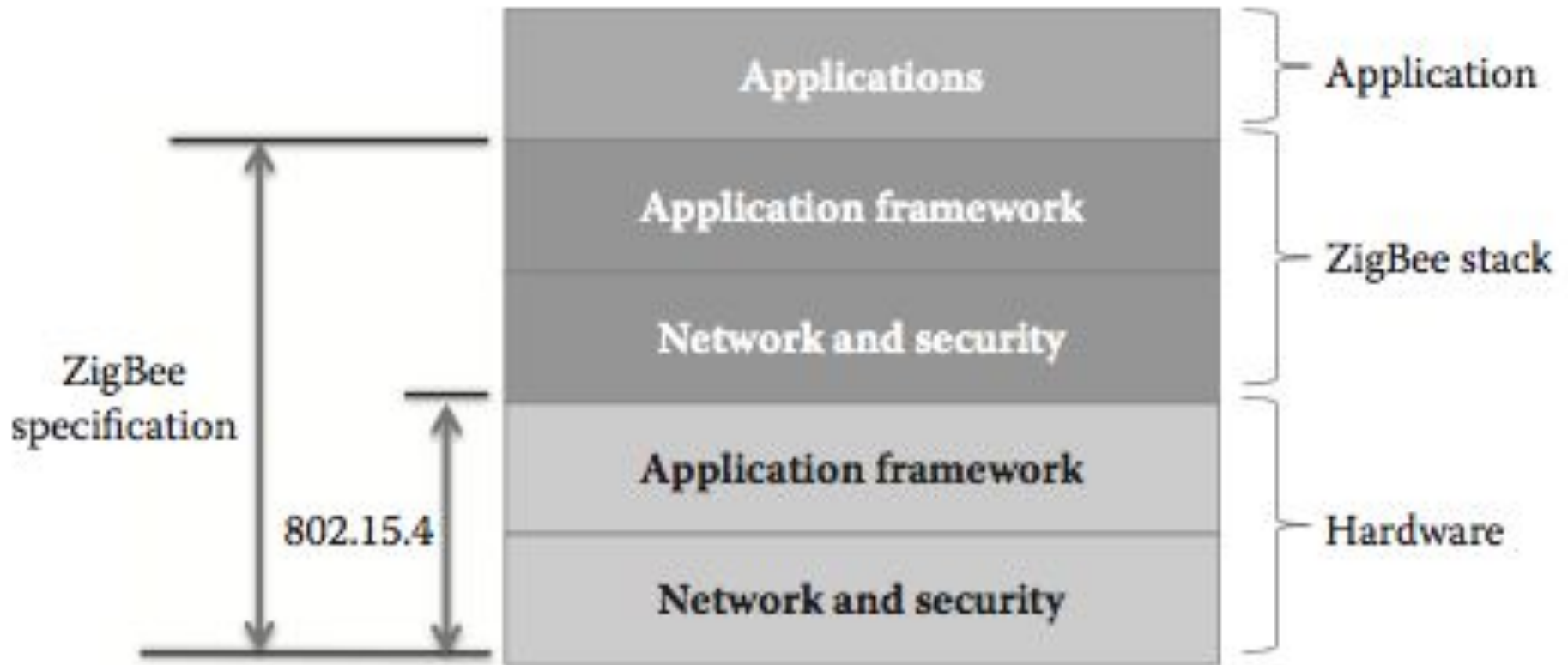
Key Use Cases of LTE

<i>LTE Service Offering</i>	<i>Usage for Intelligent Cities</i>
Live event streaming	Live coverage of key events happening in a city such as sports, concerts, award ceremonies, elections, and so on.
Real-time TV streaming	Real-time delivery of important sports events, news channels, and other popular TV shows. This will enable entertainment amid work that will in turn go a long way in boosting the productivity. In contrast to a situation where an employee may be prompted to take a leave of absence or abstain from work in order to watch some key TV event.
News, stock market reports, weather, and sports updates	Provides news, stock market reports, weather, and sports updates several times during the course of a day with on-device caching features.

ZigBee

- ZigBee protocol was framed by the ZigBee alliance. Following features of ZigBee make it very suitable for IoT applications:
 - ■ Lowpowerconsumption
 - ■ Lowcost
 - ■ Support for large number of network nodes ($\leq 65K$ nodes)

Protocol architecture of ZigBee.



Main application areas of ZigBee

Industrial and commercial

- Monitors
- Sensors
- Automation
- Control

Consumer electronics

- TV/VCR
- DVD
- Remote control

Personal health care

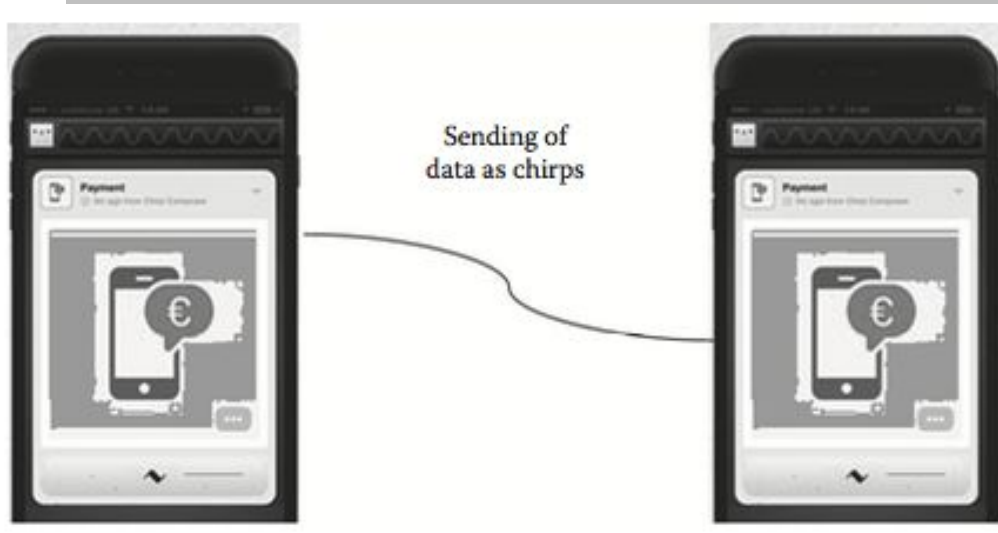
- Monitors
- Diagnostics
- Sensors

Toys and games

PC and peripherals

Home automation

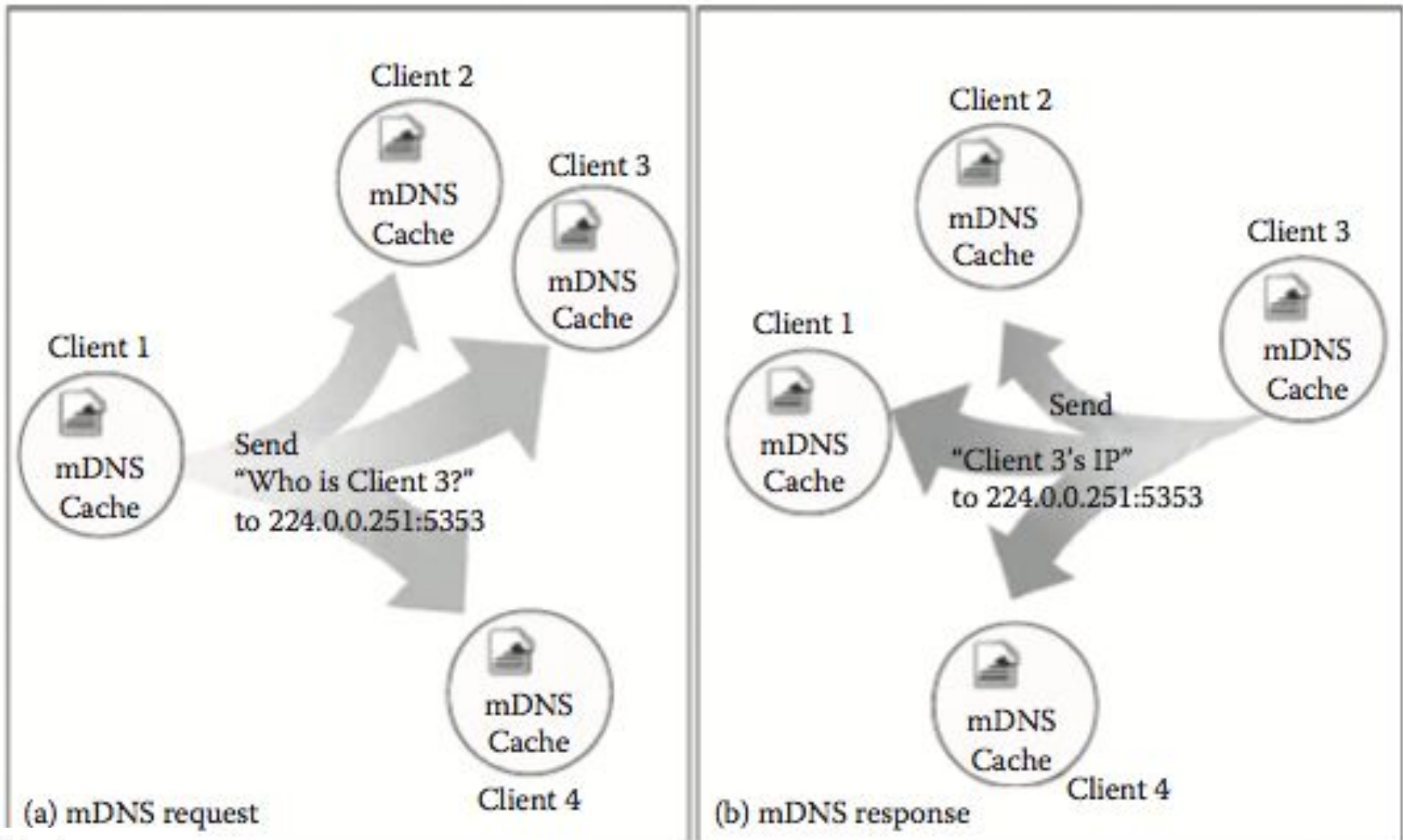
Working of chirp.



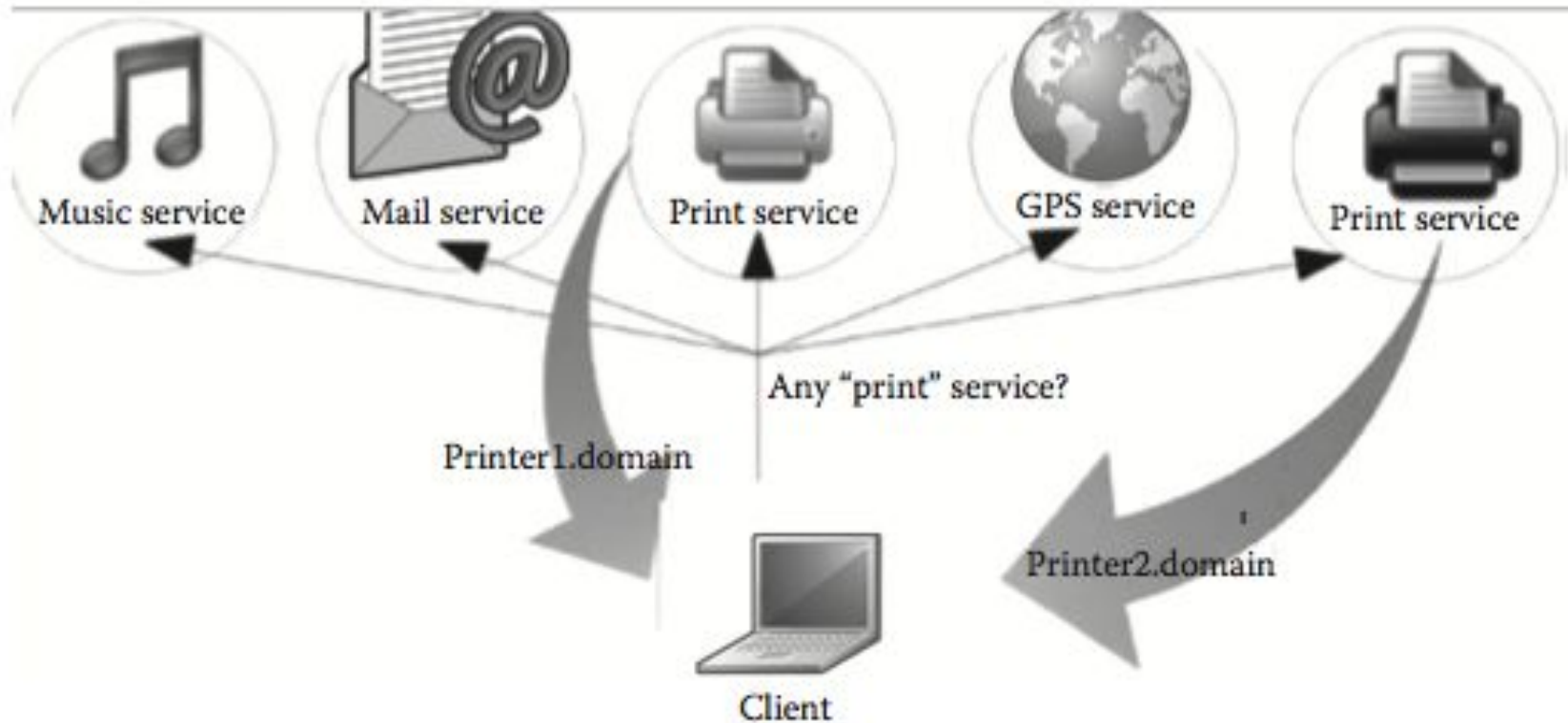
How Chirp Works

- Chirp contains a short string of data, which has been encoded using alphabets of electronic birdsong that includes 32 semitone pitches.
- Encoding and decoding of data are done locally by each mobile device, and hence it is not necessary to have an Internet connection for basic exchange. What is required is just a speaker to send data and a microphone to receive data.

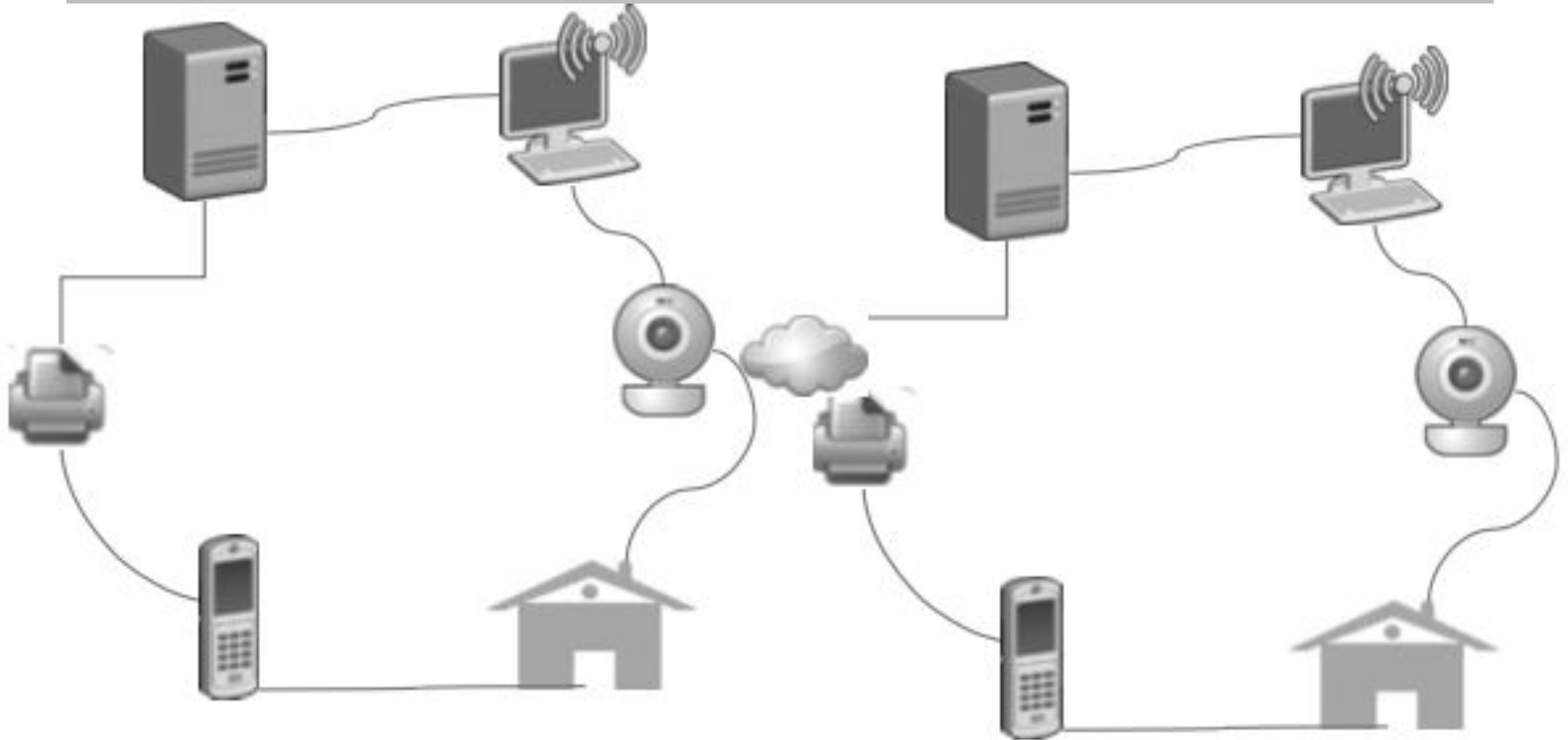
Working of multicast domain name system (mDNS)



Service discovery of a printer service using DNS-SD protocol.



Working of UPnP.



Universal Plug and Play (UPnP) is a collection of networking protocols that was devised by UPnP forum.

- The three basic components of a UPnP network are the following:
- - Devices
 - Services
 - Controlpoints
- – *Device*: It is a container for services and other nested devices, which are part of the network. A service basically is the most granular unit of control that offers a set of actions.
- – *Control points*: is provides the feature of device discovery and control by receiving device and service descriptions and by invoking service actions.
- – *Services*: e set of services that are offered by UPnP devices.